

Algebra 2 [ADVANCED]

Unit 6 Day 3: Graphical Verification & Analysis

Name: _____

Date: _____

The Goal

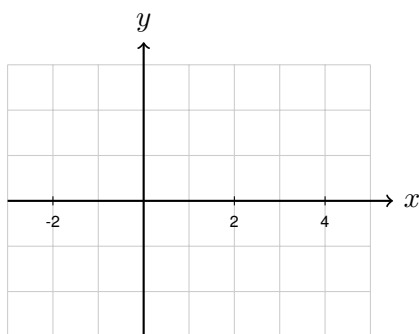
Yesterday, you solved complex equations algebraically. Today, we verify those solutions **graphically**. Remember: The solution to $f(x) = g(x)$ is the x -coordinate where the two graphs intersect.

Part 1: Verifying Non-Factorable Solutions

Refer to your Day 2 Worksheet, Problems #19 and #20.

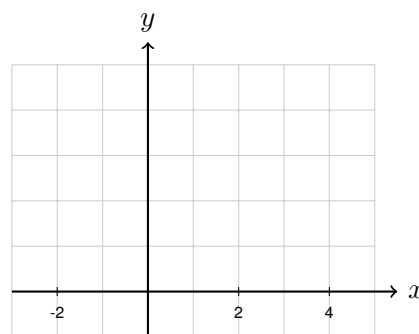
1. Analyze #19: $(x^2 - 2x + 1)^{1/3} = (2x - 1)^{1/3}$

- Algebraic solution: $x = 2 \pm \sqrt{2} \approx$ _____
- Sketch $y_1 = \sqrt[3]{(x-1)^2}$ and $y_2 = \sqrt[3]{2x-1}$.



2. Analyze #20: $\sqrt[3]{9x^2 - 7x + 2} = \sqrt[3]{-2x + 12}$

- Algebraic solution: $x \approx$ _____ and _____
- Sketch $y_1 = \sqrt[3]{9x^2 - 7x + 2}$ and $y_2 = \sqrt[3]{-2x + 12}$.



Part 2: The Mystery of the "Extraneous" Square vs. Cube

3. Solve algebraically: $\sqrt[3]{x-5} = -2$

4. Solve algebraically: $\sqrt{x-5} = -2$

5. The Comparison: Sketch both pairs. Why does the cube root have a solution while the square root does not?

Graph A: $y = \sqrt[3]{x-5}$ and $y = -2$



Graph B: $y = \sqrt{x-5}$ and $y = -2$



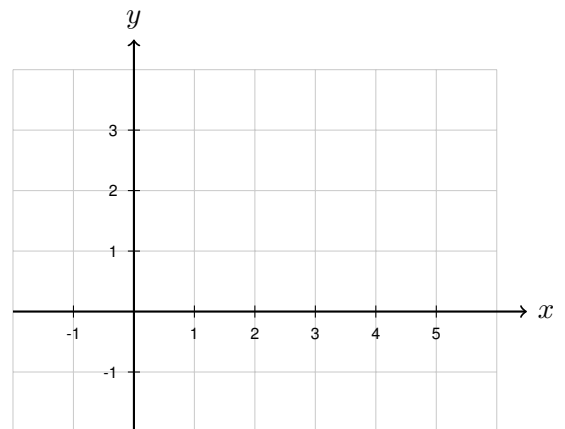
Part 4: Inequalities (The Next Level)

6. Use your graph from Problem #1 to solve: $(x^2 - 2x + 1)^{1/3} \leq (2x - 1)^{1/3}$

- Look at your sketch. On what interval of x is the "curved" graph **below** the "line-like" graph?

• **Solution** **(Interval** **Notation):**

Graph for Inequality Analysis



Part 5: Verify Your Answers

Now go back and check the rest of your answers by graphing and using the INTERSECT feature on the calculator

Exit Ticket

Explain why $x^{7/8} = 128$ only has **one** solution, while $x^{2/3} = 16$ has **two**. (Think about the numerator and denominator of the rational exponent).